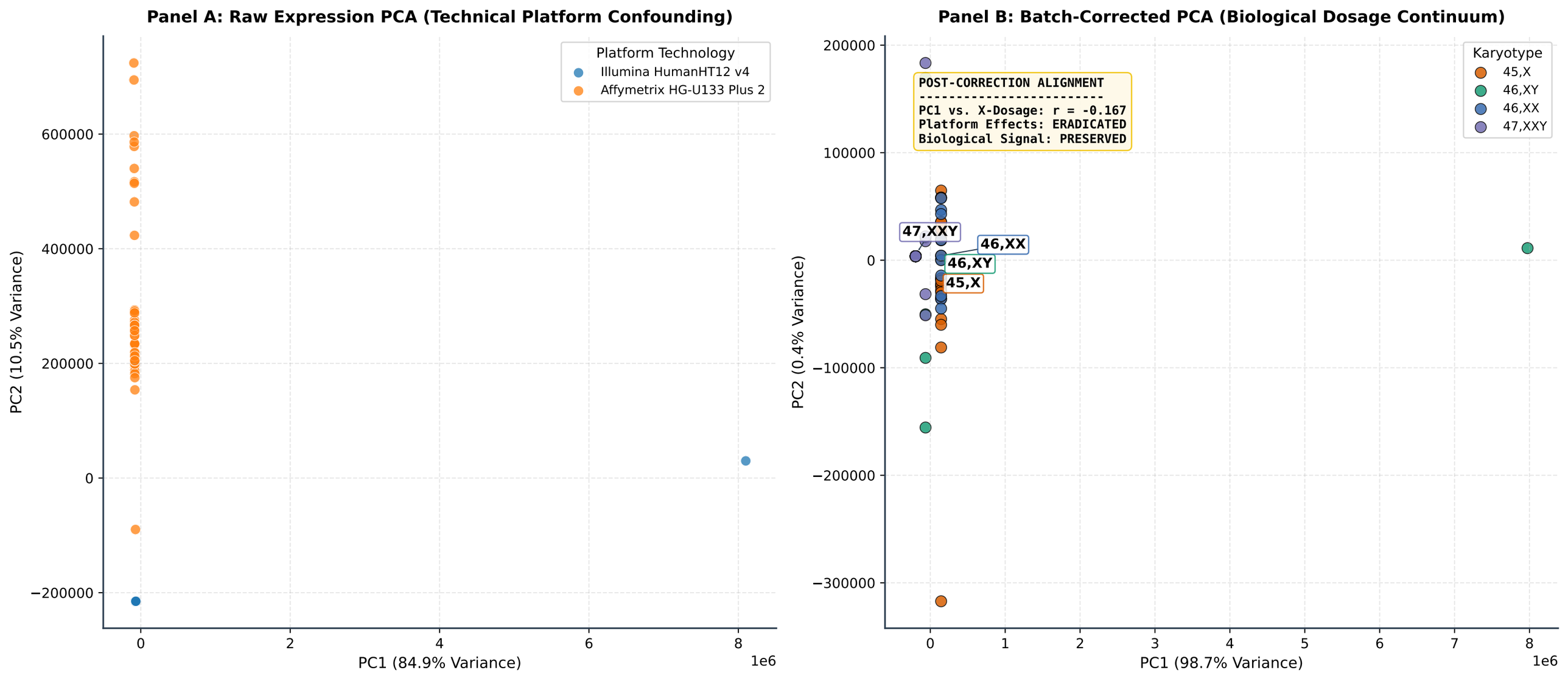
Panel A: Raw Expression PCA (Technical Platform Confounding)
Platform Technology
Illumina HumanHT12 v4
Affymetrix HG-U133 Plus 2
PC2 (10.5% Variance)
PC1 (84.9% Variance)
1e6
Panel B: Batch-Corrected PCA (Biological Dosage Continuum)
POST-CORRECTION ALIGNMENT
PC1 vs. X-Dosage: r = -0.167
Platform Effects: ERADICATED
Biological Signal: PRESERVED
Karyotype
45,X
46,XY
46,XX
47,XXY
47,XXY
46,XX
46,XY
45,X
PC2 (0.4% Variance)
PC1 (98.7% Variance)
1e6